

'Push the Industrial Complexity Away': A Vision for using Data Collection and Mixed Reality as an Analysis Tool in Industrial Product Co-Design

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Figure 1: Scenario illustrating the use of data collection and Mixed Reality (MR) for industrial product co-design. Assets obtained from freepik.com.

ABSTRACT

The rapid changes experienced in the industrial development processes to ensure interconnectivity and customization have led to an increasing need to adopt collaborative design practices and consider end-users' context at earliest stages. Such contextual information generates large amounts of data that are usually oversimplistic, overlooking the spatial dimension in its representation and visualization. Usually, this is analyzed in post-processing using digital tools, which lack proper methods to analyze the information collected. This position paper proposes a vision for using data collection and Mixed Reality (MR), thus supporting industrial product co-design. This combination can be highly flexible in providing a perspective that overlays user feedback in a digital form on top of a physical artifact. As such, it may be possible to conduct co-design tasks in a more informed manner, having a higher level of awareness of the product characteristics. These and other arguments are presented and future directions are proposed.

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1 INTRODUCTION

Central to the field of industrial design, co-designing and making sense of use patterns relative to tangible artifacts is key to conveying ideas that go beyond visualizing user's interaction over a display. Although previous research has covered the importance of co-design in graphical-user interfaces [18, 22, 27, 28], the use of such processes in assessing the user's context and interaction with a physical artifact has been overlooked. In addition, the use of mixed reality as an analysis tool to facilitate this process has been understudied.

Indeed, creating a product is a complex endeavor, incorporating several phases such as Conception, Design, and Development [3] all informed by the overarching contribution of evaluation. Before a certain product can be enjoyed by its intended users, it is submitted to various iterations during the design stage of the product development life cycle. This critical process implies a collaborative effort that accommodates unique complementary experiences and capabilities of a multidisciplinary team, working for a common purpose. Hence, it requires considering different objectives, perspectives, requirements, and priorities [18, 22, 27, 28].

Considering the rise of Industry 4.0 and its highly competitive paradigm [1, 17, 23, 26], more and more smart technologies are being introduced to reduce product development time while seeking to

decrease costs and environmental impact, as well as increase quality and flexibility [6, 12, 19]. Furthermore, as new more complex products emerge and these attend the user's context, the evaluation process conducted during the design and development phases also get more complex (e.g., generating large amounts of data for post-processing and analysis) [3]. Traditionally, the evaluation is performed using physical mock-ups, aiming to assess if the characteristics (e.g., physical design, usability, etc.) of a given product match the requirements previously established [21].

To overcome the complexity of the design process, several emerging technologies have been proposed for assisting with co-design tasks, forcing traditional design tools to be replaced by innovative digital solutions capable of sharing distinct mental representations and supporting the decision-making process of various individuals [24, 27]. Nevertheless, there is a disconnection between physical mock-ups, the digital information it was inspired by, and the data from the evaluation process, which may limit team-members ability to generate important insights [27]. In turn, this hampers the team-members capacity to make use of the plethora of data collected and generate new insights that can impact the design process.

This position paper describes a vision for combining data collection and Mixed Reality (MR) during the evaluation phase of a given product, thus contributing to improving how industrial product co-design is conducted. Using MR enables merging the advantages of virtual environments and seamless interaction with tangible objects and other individuals [5, 8, 15, 25].

Instead of just using MR to visualize a representation of the ideal design superimposed on top of the product being considered and analyzing the data collected during product evaluation afterward without direct context (as most works do according to literature [3, 4, 19, 28]), we argue that MR can also be used to foster in-context analysis of the collected data by establishing a relation between product coordinates and the data pertaining its use. In other words, a typical scenario would consider various individuals co-located or physically distributed, sharing the same MR environment which may be used to assess multiple alternatives and perfect the product design using data collected during evaluation, as represented in Figure 1.

As such, individuals can combine immediate feedback and content spreadable within physical and digital spaces to analyze the characteristics of selected products, combining the advantages of physical and virtual prototyping. By doing so, individuals may analyze the outcomes obtained during the evaluation of a product in a more informed manner. In turn, this collaborative landscape will encourage teams to communicate, interact with tangible artifacts and produce several ideas, improving customization until launch.

2 SUPPORTING MR-BASED PRODUCT CO-DESIGN AND EVALUATION

When assessing the potential of mixed-reality as an analysis tool and supporting the industrial product co-design, there are a number of arguments that can be enlisted. Some of the most prominent arguments that support our position are presented below:

- Allow designers to explore concepts/ideas/assumptions more easily than with a traditional approach;
- Enrich the way that designers experience and analyze data collected during the product design cycle;
- Increase team-members interaction with the data collected during product evaluations session on top of the real-world environment;
- Enhance co-design environments through elements from both the real and virtual worlds;
- Improves usability and credibility of the design process through a more intuitive and natural interaction approach;

- Facilitate the communication between the designers, target-users, among other stakeholders, while analyzing the data collected in an MR mode;
- Support enhanced information flow management that can significantly improve MR-based co-design productivity;
- Enable sharing information between designers during the analysis process to shorten the time for the product design life-cycle and improve quality and efficiency;
- Include designers/stakeholders/target-users empathy, contributing to raising awareness while conducting product improvement;
- Provide a platform for inspecting virtual content superimposed on the real world, boosting the creative process;
- Elicit a higher level of spatial awareness and perception during co-design tasks, as well as examine large amounts of data in a reasonable time frame.

3 DIAGRAMMATIC PROPOSAL FOR DATA COLLECTION AND MR

The diagram described in the present section (illustrated by Figure 2) brings forward a set of stages that represent the combination of data collection and MR to assist various individuals during industrial product co-design processes. As can be observed, the diagram is composed of various modules.

Considering that our proposal is intended to be used after the evaluation phase of a given physical product, our proposal for the model is intended to be used after the evaluation phase of a given product. Hence, the first module of our model is the **Product Evaluation**, which implies having a team focused on assessing specific aspects of a given product.

Current products are becoming more complex to handle the emerging needs of societies world-wide. In this vein, the **Data Collection** module ensures that large amounts of data may be obtained in a synchronous mode, allowing to create a raw data-set including a plethora of characteristics that will be analyzed later on [20] (e.g., user perception, interest, mental load, gaze, physiological variables or emotions. Also, annotations, physical mock-ups, scales, and observation grids, among others).

Then, the **Data Processing** and **Data Aggregation** modules certify raw data is expressed in a summary form (i.e., properly structured) for efficient access and analysis purposes. In turn, gaining insights about particular aspects of the product that must be addressed, while dramatically reducing query time, in particular, for large amounts of data.

Another important module is the **Data Filtering and Event Detection**. This must guarantee the possibility of selecting a set of variables for analysis. Considering that data may be collected from multiple sources and in substantial quantities, analysing it can be challenging and even overwhelming. To reduce the chances of information overload, it is imperative to ensure that designers have the freedom to customize their co-designing experience. For example, consider a partial data-set based on a pre-established list of priorities on what should be addressed first to improve certain characteristics of the product (e.g., allow to visualize using patterns from a given pilot study using a timeline approach).

Given that team-members perceive various kinds of sensory information from the real world (e.g., vision, sounds, etc.) during the evaluation stage, and that this usually is analyzed in an isolated manner, we argue that it is fundamental to focus on creating models based on different types of modalities (e.g., image, text, speech, numerical data, etc.), so that team-members have a more reliable understanding of the product being considered. As such, we propose

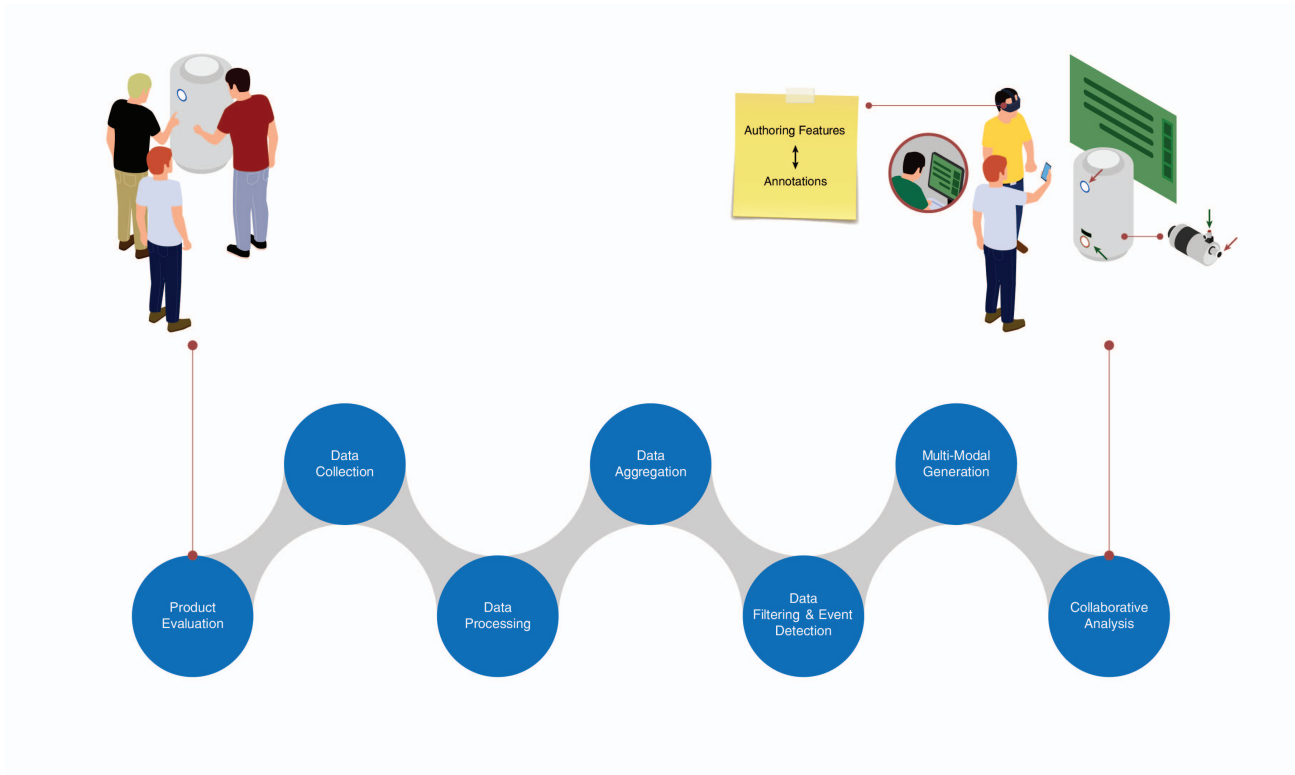


Figure 2: Diagram illustrating the necessary elements for supporting the use of data collection and MR for industrial product co-design. Assets obtained from freepik.com.

the **Multi-modal Generation** module, which is responsible for creating shared representations from arbitrary modalities. To elaborate, typically, the data collected during the evaluation process lacks synchronization and context, hindering designers from understanding the collected data as a whole. Therefore, this module focuses on giving meaning to the data collected, by preparing this information to be put into perspective using MR visualizations features (e.g., merging different types of coordinates, enabling using patterns to be better perceived in relation to their context by the team-members) in the next module. In other words, creating a unified representation of the data collected. This vision promotes a shared interaction mechanism down the pipeline, by merging data in a spatial manner and in relation to the physical product, with all that this implies. To contextualize a bit further, using eye gaze may help provide a clue on what was the focus of the users that evaluated a given product, as well as if a given problem that may have occurred was caused by the product, or for a third-party entity in the surrounding context (e.g., understanding that the cause of a problem on a heating equipment in a smart house was not the equipment itself, but a gas connection that affected the evaluation test of a given member of a family - this way, what could initially have been perceived as a defect of the product, could be clarified. The same principle can be applied to other examples, and with other data types highlighting important changes that need to occur).

As a result, a co-located **Collaborative Analysis** session, may occur among various team members (while also including remote members [9, 11, 13]), taking advantage of the pipeline previously described. In this stage, using distinct MR devices (e.g., Hand-Held Device (HHD), Head-Mounted Display (HMD)), team-members may preview different types of data on top of the real-world product, thus presenting situated visualizations [2, 7, 10, 16] (e.g., visualiz-

ing heat maps resulting from the eye gaze collected, consequently illustrating components that still require further iterations). This may be conducted simultaneously for all data types or for selected variables. Furthermore, through authoring capabilities, distinct types of annotations (i.e, using digital assets superimposed on the physical product like text, predefined shapes, drawings, or others [12, 14, 15]) may be used to suggest adjustments and improvements. By doing so, team-members can interact with and modify this virtual content, contributing to better expressing themselves, sharing ideas and assumptions, as well as previewing possible updates before they are approved for mass-production (e.g., removal, reorientation, or repositioning of components, resulting in new designs). Moreover, by using various supporting devices, a different spectrum of interactions may be provided according to the platform being used. To finish, all results of the co-design process may be updated to a cloud database, so that all content can be re-visited if necessary, or edited down the line.

4 FUTURE DIRECTIONS

With the goal of exploring the impact of using data collection and MR for industrial product design, there are still some topics that must be addressed by the research community, namely:

- **Monitoring Framework** - allow data collection regarding various measurements, which must later be made available for consumption through MR;
- **Visualization** - generate an interactive data visualization that utilizes coordinates from the physical product to display the gathered data, allowing real-time exploration without losing context;

- **Interaction** - explore additional modalities for creating and exploring virtual content in a flexible and timely manner;
- **Authoring features** - support the creation of different types of virtual content (e.g., text, drawings, pre-defined shapes, images, etc.);
- **Multi-platform** - ensure different devices can be supported, according to the environment/task context. In fact, distinct devices should be used simultaneously;
- **Multi-user collaboration** - provide the possibility of using MR in a distributed manner (either in a co-located or remote setting), allowing more than one person to view and analyze the data collected, as well as suggest changes to the product. Plus, include the capacity to verify co-design history, and avoid editing conflicts, among others.

Many aspects of these topics remain unanswered, i.e., extensive research effort in multiple disciplines is still required. In our understanding, future work should seek to determine the extent to which MR can be used to assist industry product design, taking advantage of existing data collected during evaluation processes. Also, further work should confirm if there are observable differences in the use of different visualization and interaction methods under such circumstances.

5 CONCLUDING REMARKS

To sum up, the authors strongly believe that the evaluation process of such products needs to evolve to embrace their complexity and evaluation scenarios require to go beyond the lab. From this perspective, our team proposes the adoption of MR technologies for visualizing and interacting with the immensity of data collected during the evaluation of any given physical product. By doing so, MR gains new value, becoming a novel tool capable of supporting teams focused on product design in industrial scenarios.

Due to its characteristics, various types of measurements may be overlapped on top of the real-world environment, providing a different perspective from which it is possible to analyze, discuss and generate new ideas for improving how certain products must be tweaked before going to mass-production. Not only that, but MR technology can also be applied to transform team-members ideas into tangible artifacts that can help advise how a product needs to be perfected with a higher level of detail and precision, e.g., using spatial notes to explain a specific alteration that needs to occur.

In this vein, we argue that an opportunity exists to explore how MR technologies can assist in such contexts, in particular, taking the ideas from the proposed diagram to the next level, i.e., designing and developing a framework capable of supporting the ideas considered, as well as applying the framework to a real-life example in order to test its capacity to handle a realistic scenario over various iterations with a team(s) of multidisciplinary individuals.

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